

Adnan Ahmed Khan

Senior Backend & Platform Engineer | Node.js, Python, AWS, Distributed Systems
Islamabad, Pakistan | Remote Friendly
khanadnanahmed01@gmail.com | linkedin.com/in/adnahmed | github.com/adnahmed

SUMMARY

Senior Backend & Platform Engineer with 6+ years building high-throughput, business-critical platforms. Scaled systems to 500–800 RPS and 50k–100k DAU while cutting AWS spend 25%. Slashed 30s+ queries to < 1s, MTTR from ~45m to < 10m, and failed ops by 80%.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python

Backend: Node.js, Nest.js, FastAPI, Express, REST, workflow orchestration

Cloud/Infra: AWS, EKS, ECS, Lambda, Docker, Kubernetes, Terraform, Helm

Data/Messaging: Postgres, Redis, Kafka, Elasticsearch, MongoDB, DynamoDB

Architecture: Microservices, distributed systems, event-driven design, idempotency, retries

DevOps/SRE: GitHub Actions, ArgoCD, Prometheus, Grafana, Jaeger, Datadog, ELK, Jest

PROFESSIONAL EXPERIENCE

HackOnTech

2023 – Present

Senior Software Engineer, Backend & Platform

Remote / Islamabad

Core backend owner for B2B/G2G NGO platforms: 50k–100k DAU, 500–800 RPS peak.

- Spearheaded 0 → 1 platform overhaul into 6 high-throughput microservices, enabling zero-downtime releases for 100k DAU.
- Cut failed cross-service operations 80%, reducing daily failures from 150 to < 30 across critical workflows.
- Stabilized 500–800 RPS burst windows while protecting payments, queues, and customer-facing availability.
- Slashed core query latency from 30s+ to < 1s across high-volume Postgres/Redis workloads.
- Reduced payment failures from ~5% to 2% with resilient retries, rollbacks, and workflow recovery.
- Reclaimed 25% of AWS spend while preserving SLOs across EKS, autoscaling, and non-prod workloads.
- Cut MTTR from ~45m to < 10m with tracing, dashboards, and service-level observability.

Stack: TypeScript, Node.js, Nest.js, Python, FastAPI, AWS, EKS, Lambda, Postgres, Redis, Temporal, Step Functions, Terraform, Helm, ArgoCD, Prometheus, Grafana, Jaeger.

Open-Source & Production-Grade Engineering Platforms

Distributed systems, agent runtimes, rendering infrastructure, collaboration engines

- Architected verifiable-browser-agent, an async browser-agent runtime with LangGraph orchestration, grounded Playwright actions, semantic effect verification, and distributed recovery.
- Architected Vidx, a 4K distributed rendering platform, reducing peak memory 40% under GPU-bound burst loads.
- Designed pdf-annotate.js collaboration engine with conflict-safe sync, spatial indexing, and 60fps rendering.
- Engineered openclaw-workflow orchestration runtime with dependency graphs, resumable execution, retries, and output gates.

Stack: Python, FastAPI, LangGraph, Playwright, Node.js, TypeScript, Redis, WebSockets, FFmpeg, GPU workers, queues, PDF rendering, distributed execution.

Creative IT Park

2021 – 2023

Backend Developer

- Delivered backend platforms for university, government, and healthcare systems with production API ownership.
- Hardened Kafka notification pipelines, preserving delivery during downstream service failures.
- Cut search latency across public portals using optimized data access and caching.
- Secured healthcare ETL pipelines with encrypted flows and fine-grained AWS access controls.

Stack: Node.js, Python, FastAPI, Kafka, Elasticsearch, Redis, AWS ECS, RDS, DynamoDB, Glue, IAM, PostgreSQL, MongoDB, Docker.

EARLIER EXPERIENCE

GeekGenix – Backend Developer | 2020 – 2021: Stripe checkout, webhook reconciliation, Express/MongoDB APIs, Passport.js auth.

EDUCATION

BS Computer Science — PMAS Arid Agriculture University, Rawalpindi